Cells Under the Microscope

Biology 2e chapter 4 reading connection | Teacher guide / classroom preparation

Learning target

How do the visible structures and sizes of different cells compare?

Timing and materials

One class period; prepare slides and calibrate equipment beforehand.

Compound microscope, slides and coverslips, dropper, water, onion epidermis, school-approved aquatic leaf specimen, prepared animal epithelial slide, stage micrometer and eyepiece reticle.

Classroom route

1. Make thin plant wet mounts using a small drop of water. Use a prepared animal slide; do not collect human specimens. Lower coverslips at an angle and handle glass carefully.

2. Find the specimen at low power. Calibrate the eyepiece reticle against a stage micrometer separately for each objective, then use fine focus at higher power.

3. Measure two non-overlapping fields per specimen, with a consistently selected cell in each. Record the calibration, length, visible structures, and uncertainty. More cells improve the sample.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

Students should apply the objective-specific calibration and acknowledge sampling uncertainty. Look for evidence of cell walls in plant schematics and chloroplasts in the aquatic leaf. Onion bulb epidermis generally lacks visible chloroplasts. Do not grade real cells against simulated sizes.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Manual tallies are student observations; a matching total does not prove every category is correct. Do not replace student counts with the model answer. The optional Sordaria extension retains classification feedback.

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Model limitations

Original schematic cells, not micrographs. Dimensions illustrate plausible ranges, not species measurements. Optical resolution, staining and specimen thickness are simplified.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Corn and fly activities require manual tallies summing to the sample size plus review; optional specimen marks do not classify or tally automatically. Chromosome models require all four gametes plus a reviewed tally; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

Plant cell microscopy

https://www.carolina.com/teacher-resources/Document/carolina-labsheets-observing-plant-cells/tr38030.tr

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.